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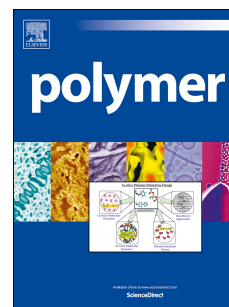
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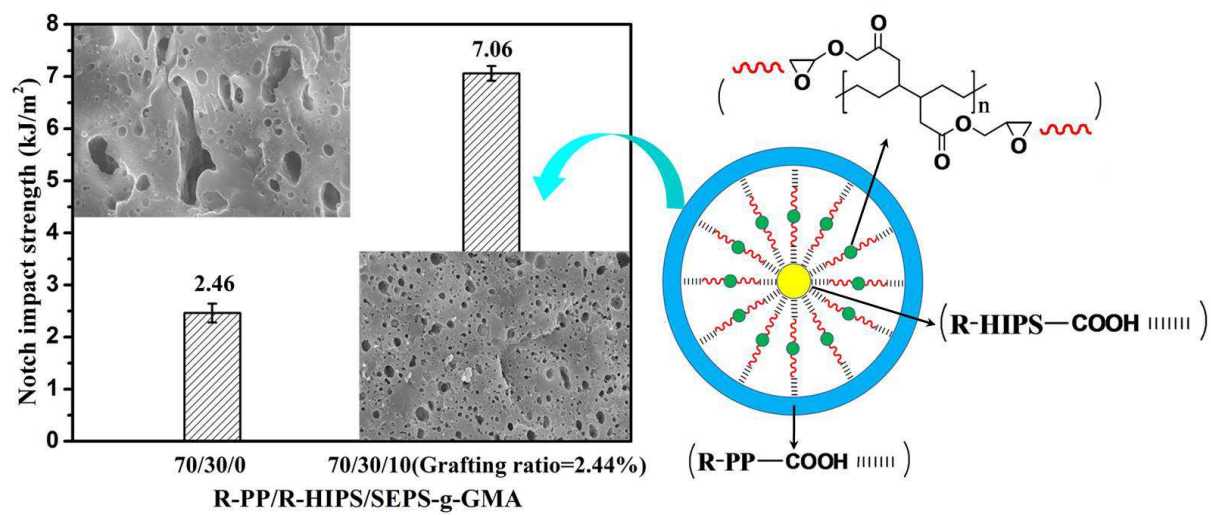
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Graphical Abstract



Preparation of Polystyrene-b-poly(ethylene/propylene)-b-polystyrene Grafted
Glycidyl Methacrylate and its Compatibility with Recycled
Polypropylene/Recycled High Impact Polystyrene Blends

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Abstract: In this study, polystyrene-*b*-poly(ethylene/propylene)-*b*-polystyrene grafted glycidyl methacrylate (SEPS-*g*-GMA) copolymer was prepared by melt grafting in the mixer. The successful grafting of GMA on the SEPS molecular chain was analyzed by Fourier transform infrared spectroscopy (FT-IR). The grafting ratio of SEPS-*g*-GMA was measured by acid-base titration, which showed that N-vinylpyrrolidone (NVP) had a positive effect on the increased grafting ratio of SEPS-*g*-GMA. The recycled polypropylene (R-PP)/recycled high impact polystyrene (R-HIPS) blends were prepared by a melt extrusion, and the effect of SEPS-*g*-GMA copolymer with different ratios on the compatibility of blends was studied. The epoxy groups in SEPS-*g*-GMA copolymer were found to have a chemical reaction with the carboxyl groups in the waste materials. When the grafting ratio of SEPS-*g*-GMA was 2.44%, the notched impact strength and the elongation at break of the R-PP/R-HIPS blends with 10 phr SEPS-*g*-GMA reached 7.06 kJ/m² and 34.25%, which were significantly increased by 186.99% and 68.30% compared with that of pure blends, respectively. Moreover, the observed decreased particle size and increased dispersion uniformity improved the compatibility using SEPS-*g*-GMA. The increased complex viscosity, storage modulus and loss modulus indicated that the chemical reaction between SEPS-*g*-GMA and R-PP/R-HIPS blends improved the component compatibility among the blends, resulting in the chain entanglement prominently. The presence of SEPS-*g*-GMA inhibited the degradation and increased the thermal stability of R-PP/R-HIPS blends.

Keywords: Recycled polypropylene; Recycled high impact polystyrene; Compatibilization.

1. Introduction

With the rapid development of economy and the improvement of people's living standard, the use time of electronic and electrical equipment (EEE) becomes shortened, and the speed of upgrading is accelerating in modern society [1, 2]. Thus, a large number of EEE waste (WEEE) will also be produced. It was reported that 20-50 million tons of WEEE are generated every year around the world before 2012. A continuous growth trend had been also maintained, which was expected to increase by 200 to 500 percent by 2020 [3-5]. A variety of recyclable and reusable resources exists in these WEEE, including non-ferrous metals, plastics and so on. The plastic content accounts for about 20 percent to 30 percent, all of which are basically from the shells of WEEE [6, 7]. These waste plastic shells mainly contain polypropylene (PP) and high impact polystyrene (HIPS) materials [8-10]. The way to deal with them include simple melt extrusion and granulation, or using thermal energy by cracking [11, 12]. These methods could not play the greatest value of materials, because most of these materials still have some main properties and potential of modification and reuse [13].

We know that PP has good heatproof and corrosion resistance, and HIPS has outstanding dimensional stability and electrical insulation. The performance would be not lost with the aging of the materials in a long period of use [14, 15]. If the advantages of the recycled PP (R-PP) and the recycled HIPS (R-HIPS) were combined to improve the mechanical properties which would decrease the aging degradation process. Meanwhile, by mixing, not only composite materials with excellent comprehensive properties can be prepared, but also the conservation of fossil fuels

and the control of environmental pollution can be minimized significantly [16, 17].

Due to the active tertiary carbon atoms on the PP molecular chain and the unstable C=C in the molecular chain of polybutadiene in HIPS, PP and HIPS will undergo aging degradation after being used for a long time, resulting in the fracture of polymer chains and the formation of functional groups such as hydroxyl, carbonyl and carboxyl groups [18-21]. It is reported that the compatibilization of virginal PP and HIPS is mainly modified with block or graft copolymers [22, 23]. We have also used polystyrene-*b*-poly(ethylene/propylene) (SEP) and polystyrene-*b*-poly(ethylene/propylene)-*b*-polystyrene (SEPS) to improve the physical compatibilization of R-PP/R-HIPS blends in previous studies, which exhibited a significant improvement. However, these methods could not recover the fractured molecular chain and caused aging and degradation. If a reagent can be found to react with these functional groups generated by aging to improve the molecular weight and chemical compatibility between R-PP and R-HIPS, the R-PP/R-HIPS composites will possess more excellent performance [18].

Glycidyl methacrylate (GMA) is a kind of functional monomer. Its active vinyl and epoxy groups could be grafted with polyolefin and reacted with the polar groups such as amine, hydroxyl and carboxyl group, respectively [24, 25]. Therefore, the fusion grafting method was applied to graft GMA on the SEPS macromolecule chains to form a functional polystyrene-*b*-poly(ethylene/propylene)-*b*-polystyrene graft glycidyl methacrylate (SEPS-*g*-GMA) copolymer, which was used as a compatibilizer to improve the compatibility of R-PP/R-HIPS blends in this study. However, the grafting ratio and grafting efficiency of GMA were not high, because of the high homopolymerization ability and the low reactivity with

polyolefin molecular chains. Styrene (St) monomer is often used as a co-monomer of GMA grafted on polyolefin, which can greatly improve the grafting ratio and grafting efficiency [26, 27]. However, N-vinylpyrrolidone (NVP) was chosen as the co-monomer to promote the increase of GMA grafting ratio and grafting efficiency in this experiment, which is ascribed to its higher reactivity and efficiency than St in the preparation process of elastomeric graft GMA copolymer [28, 29].

In this work, SEPS-g-GMA copolymer was successfully prepared and the effects of NVP on the grafting ratio of SEPS-g-GMA were studied by acid-base titration. Moreover, SEPS-g-GMA was added into the R-PP/R-HIPS blends as a compatibilizer. The physical and chemical compatibilization effects of SEPS-g-GMA on the blends were studied by Fourier transform infrared spectroscopy (FT-IR), mechanical properties, morphology, rheological properties and thermogravimetric analysis (TGA). The mechanisms were explored to illustrate the compatibility role of SEPS-g-GMA in the recycled PP and HIPS.

2. Experiment

2.1 Materials

The R-PP and R-HIPS, from WEEE shell plastic, were provided by Shunde Xinhuanbao Environmental Protection Materials Co., Ltd (FoShan, China). By using the crusher, the plastics were broken into fragments of 5 millimeters in diameter for blending modification. The SEPS copolymer with 20% polystyrene block (Kraton G-1730) was provided by Shanghai Jingying Material Co., Ltd (Shanghai, China). The GMA (Industrial grade) was purchased from Tianjin

Huolong Electronic Materials Co., Ltd (Tianjin, China). The NVP (Industrial grade) was supplied from Jining Huakai Resin Co., Ltd (Jining, China). The dicumyl peroxide (DCP, Chemical pure) was provided by Shanghai Lingfeng chemical reagents Co., Ltd (Shanghai, China). Toluene, xylene, acetone, ethanol and tetrahydrofuran were purchased from Tianjin Guangfu Fine Chemical Research Institute Co., Ltd (Tianjin, China).

2.2 Preparation of SEPS-g-GMA

The SEPS was added to the preheated internal mixer (HL-200, Suyan Science and Technology Co., Ltd, China) with the rotor speed of 40 rpm at 200 °C for 3 min. The mixed solution of NVP, GMA and DCP in a certain proportion was carried out in the SEPS melt and the SEPS-g-GMA copolymer was prepared after 5-min reaction. The specific formula was implemented according to Table 1 and the synthesis mechanism was carried out in accordance with Scheme 1.

2.3 Purification of the SEPS-g-GMA

The SEPS-g-GMA was dissolved in toluene at room temperature and then excess acetone was added to precipitate it which was filtered through a sand core funnel to remove free GMA, NVP and their polymers. The samples were washed and repeated several times in accordance with the above steps to obtain the pure SEPS-g-GMA copolymer for testing, and then being used as the compatibilization of the R-PP/R-HIPS blends.

2.4 Determination of the grafting ratio of SEPS-g-GMA

The grafting ratio (G) of SEPS-g-GMA was measured by acid-base titration in the experiment. 1 g SEPS-g-GMA, obtained above, was accurately weighed by the analytical balance and added to a flask with 100 mL toluene. Then 15 mL toluene solution of acetocastin with a concentration of 0.10 mol/L was added and heated to boiling for 1 h. Afterwards, the ethanol solution of sodium hydroxide with a concentration of 0.05 mol/L was used to titrate to the ending point (The ethanol solution of phenolphthalein was used as the indicator).

The calculation of grafting ratio was obtained by Formula (1):

$$G = \frac{(V_{\text{CCl}_3\text{COOH}} \cdot C_{\text{CCl}_3\text{COOH}} - V_{\text{NaOH}} \cdot C_{\text{NaOH}}) \cdot M_{\text{GMA}}}{W_s} \quad (1)$$

where $V_{\text{CCl}_3\text{COOH}}$ (L) is the volume of the toluene solution of acetocastin consumed in the titration experiment, and $C_{\text{CCl}_3\text{COOH}}$ (mol/L) is the concentration of the toluene solution of acetocastin. V_{NaOH} (L) and C_{NaOH} (mol/L) represent the volume and the concentration of the sodium hydroxide ethanol solution, respectively. M_{GMA} means the molecular weight of GMA and W_s represents the quantity of the sample.

2.5 Preparation of the blends

The blends were prepared by melt blending on the basis of Table 2 in a SHJ-36 co-rotating twin screw extruder (Nanjing Chengmeng Chemical Machinery co., Ltd, China) with a screw speed of 85 rpm and a feed rate of 25 rpm. The temperatures from barrel to head were: 180°C, 185°C, 190°C, 195°C, 200°C, 205°C, 205°C, 205°C, 205°C, respectively. After cooling and granulation, the blends particles were dried for 8 hours in the dryness box and then the standard samples were obtained by the SZ-100/80 injection molding machine (Shanghai Plastic

Machinery Co., Ltd, China) with the injection temperatures were: 190°C, 195°C, 200°C, 200°C, respectively.

2.6 Fourier transform infrared spectroscopy (FT-IR)

The samples and potassium bromide (KBr) were dried and pressed into thin sheets, which were detected in the Fourier transform infrared spectrometer (Bruker Tensor 27, Bruker Optik GmbH, Spain) with a range from 500 to 4000 cm^{-1} . The location of wavenumber, the number of wave crest and the absorption intensity were analyzed.

2.7 Mechanical Testing

The impact specimens were notched with a 45° angle and relaxed for 24 hours at room temperature. The K-TEST KXJU-22A notch impact testing machine (Chengde Science Standard Instrument Testing and Manufacturing Co., Ltd, China) was used to test the notch impact strength according to GB/T 1043-2008 at room temperature with a 1-J hammer. The tensile strength was tested on the SANS CMT6104 microcomputer controlled electronic universal testing machine (Shenzhen Skyan Power Equipment Co., Ltd, China). According to GB/T 1040.1-2006, the fixture distance and tensile speed were 115 mm and 50 mm/min at room temperature, respectively. All of the tests above were tested with 5-6 samples, and the average value was reported.

2.8 Scanning Electron Microscope (SEM)

The R-PP/R-HIPS composite specimens were fractured in liquid nitrogen, and then were etched in tetrahydrofuran for 24 hours. After drying in a vacuum drying oven, the samples were

observed by the Hitachi-SU8010 (Techcomp Co., Ltd, China) field emission scanning electron microscope after spraying gold on the surface. The acceleration voltage was 10 kV.

2.9 Rheological properties

The rheological properties of R-PP/R-HIPS blends were tested by Bohlin CVO150 rotational rheometer (Malvern Instruments Co., Ltd, United Kingdom) at 180 °C. The diameter of the parallel circular plates was 25 mm, and the space between the plates was 1 mm. The strain amplitude was 1% and the scanning frequency was fluctuated from 0.01 to 100 rad/s, so as to ensure that the test was carried out in the linear viscoelastic region.

2.10 Thermogravimetric analysis (TGA)

TGA of R-PP/R-HIPS blends was measured by the Mettler Toledo TGA1 thermogravimetric analyzer (Mettler Toledo Co., Ltd, Switzerland) in nitrogen atmosphere with the flow rate being 50 mL/min. The sample was dried for 24 h at 70 °C before the test. The temperature range of the test was from room temperature to 600 °C and the heating rate was 10 °C/min. During the experiment, the thermal weight loss curve of the sample was recorded and the derivative of thermal weight loss curve was calculated.

3. Results and discussion

3.1 Characterization of the SEPS-g-GMA copolymers

Fig. 1 shows the FT-IR contrast diagram of SEPS and SEPS-g-GMA. The curve "a" and "b" correspond to the characteristic peaks of SEPS and SEPS-g-GMA, respectively. As can be

seen clearly in the figure, the absorption near 913 cm^{-1} in SEPS-g-GMA caused by the C-O expansion vibration in the epoxy group is obviously enhanced compared with the SEPS [30]. The absorption peak near 1715 cm^{-1} represents the vibrational peak of C=O in GMA [27, 31], which does not exist in pure SEPS. There are several absorption peaks in the vicinity of 1880 cm^{-1} , which were stronger than those of pure SEPS. This may be caused by the stretching vibration of C=O in NVP. Two obvious absorption peaks at around 3078 cm^{-1} are attributed to the vibration of C-H in the five membered heterocyclic ring of NVP. These results indicate that GMA has been successfully grafted onto the SEPS chain and NVP may also be grafted on the SEPS molecular chain as a co-monomer.

3.2 Grafting ratio of SEPS-g-GMA

Fig. 2 shows the effect of monomer NVP content on the graft ratio of SEPS-g-GMA. It can be seen that when the NVP content is 4 phr, the grafting ratio of SEPS-g-GMA is 1.10%. With increasing the NVP amount, the grafting ratio of SEPS-g-GMA could be increased to 5.40% when the NVP content is 7 phr. The result indicates that NVP may promote GMA grafting on the SEPS molecular chain, which is conducive to improve the grafting ratio of SEPS-g-GMA.

3.3 Characterization of the R-PP/R-HIPS/SEPS-g-GMA blends

Fig. 3 shows the changes of infrared spectrum characteristic peaks of SEPS-g-GMA as the compatibilizer to compatibilize the R-PP/R-HIPS blends. It can be seen that the absorption peak at the 1400 cm^{-1} corresponds to the single bond vibration of O-H in the carboxyl groups, and the absorption peak near 3000 cm^{-1} is due to the vibration of the carboxyl groups. The obvious

weakening of these two absorption peaks may be due to the opening of epoxy group in the SEPS-g-GMA and the reaction with the carboxyl groups in the waste plastics. The absorption peak near 1660 cm^{-1} is due to the carbonyl groups on the NVP in the grafted copolymer. The absorption peaks at 1721 and 1742 cm^{-1} correspond to the vibration of the carbonyl group in the carboxyl group and ester group, respectively. The migration of the peak position of the carbonyl group is probably due to the reaction between the epoxy groups in the graft copolymer and the carboxyl groups in the waste plastics, during which the ester functional group is produced [32]. The results of FT-IR analysis show that SEPS-g-GMA reacted with the carboxyl groups in the blends to increase the molecular chains of the blends and to improve their compatibility. The reaction mechanism is shown in Scheme 2.

3.4 Mechanical properties analysis

The effect of SEPS-g-GMA with different grafting ratios on the notch impact strength of the R-PP/R-HIPS blends can be observed from Fig. 4. The notch impact strength of R-PP/R-HIPS blends without any compatibilizer is only 2.46 kJ/m^2 , which may be attributed to the aging and degradation during a long period of using and the poor compatibility between R-PP and R-HIPS. The interruption of force during the phase transfer makes the material to be damaged easily. When 10 phr SEPS was added into the R-PP/R-HIPS blends, the notch impact strength of the blends could reached 6.46 kJ/m^2 , which was 162.60% higher than that of pure blends. The significantly increased notch impact strength is probably due to the fact that SEPS contains the poly(ethylene/propylene) block similar to R-PP and the same polystyrene block with R-HIPS,

which leads to an outstanding increase of the R-PP/R-HIPS blends compatibility [33]. Meanwhile, when the material is affected by external force, SEPS can also induce a large number of crazes in the blends, and the energy is consumed by the deformation of shear yield [34]. The impact strength of the blends continues to be enhanced with the addition of SEPS-g-GMA, and reaches a maximum value of 7.06 kJ/m^2 which is increased by 186.99% in the graft ratio of 2.44% compared to pure blends. This may be due to the reaction of the epoxy group in SEPS-g-GMA with the carboxyl group produced in the waste plastics, which can improve the compatibility of the blends from the point of chemical compatibilization and increase the molecular weight of any phase in the blends, respectively. With the increase of compatibility, the force of two interphase molecules is enhanced and the intermolecular slippage is not easy to produce. The energy caused by stress can be quickly transmitted and absorbed at the moment of high speed load impact, which makes the impact strength increase [35]. When the graft ratio of SEPS-g-GMA is more than 2.44%, the impact strength of the blends is decreased, which may be due to the fact that R-PP and R-HIPS are non-polar polymers with a small amount of carboxyl functional groups produced in aging degradation process. The SEPS-g-GMA copolymers with higher grafting ratio have a greater polarity, which may have a negative effect on the compatibility of the blends and be not conducive to improve the impact strength after the saturated reaction of the compatibilizer and the waste plastics.

The effect of SEPS-g-GMA copolymer with different grafting ratios on elongation at break of the R-PP/R-HIPS blends is exhibited in Fig. 5(a). It can be seen from the diagram that the elongation at break of the blends with 10 phr SEPS is increased by 57.06% compared with the

blends without the compatibilizer whose breaking elongation is only 20.35%. With the addition of 10 phr SEPS-g-GMA in the blends, the elongation at break increased initially and reached the maximum 34.25% when the grafting ratio of SEPS-g-GMA was 2.44%, which was 68.30% higher than that of pure R-PP/R-HIPS blend. Then the elongation at break decreased with increasing the grafting ratio. The breaking elongation of the R-PP/R-HIPS blends with the change of graft ratio of SEPS-g-GMA is similar to the change of the notch impact strength. This may be due to the synergistic effect of SEPS-g-GMA on the physical and chemical compatibilization of the blends. Meanwhile, SEPS-g-GMA may also play a role in repairing the molecular chains of each phase in the blends.

The tensile strength of the blends with varied grafting ratio of SEPS-g-GMA is different from the notch impact strength, which can be observed from Fig. 5(b). The tensile strength, mixed with 10 phr SEPS-g-GMA, decreases slightly compared with pure blends whose tensile strength is 21.88 MPa. This may be attributed to the fact that SEPS is an elastomer with a lower tensile modulus and affects the rigidity of the blends. With the addition of the same mass fraction of SEPS-g-GMA copolymer, the tensile strength of the blends increased gradually. The maximum value reached 22.91 MPa when the grafting ratio was 4.40%. This may be due to the reaction between the epoxy group in SEPS-g-GMA and the carboxyl group in the waste plastics, which improves the molecular weight of each phase and compatibility of the blends. Subsequently, the tensile strength of the blends was decreased, which may be imputed to the greater grafting ratio that gives SEPS-g-GMA a higher polarity but negatively to the compatibility of the blends.

3.5 Morphology analysis

The compatibility can be characterized by the micromorphology of the R-PP/R-HIPS blends shown in Fig. 6. The irregular holes are obtained from the R-HIPS phase in the blend after washing by tetrahydrofuran. It can be seen from the morphology of blends without compatibilizer shown in Fig. 6(a) that the size of R-HIPS particles dispersed in R-PP matrix is large, uneven and widely distributed, indicating that R-PP and R-HIPS are simply mechanical mixed with poorer cohesive force between them and thus resulting in poor miscibility. When mixed with 10 phr SEPS block copolymer, the particle size of dispersed phase decreases significantly compared with pure R-PP/R-HIPS blend, which can be observed from Fig. 6(b). Meanwhile, the dispersion of holes is also more uniform, which is attributed to similar molecular structure in the SEPS and R-PP/R-HIPS blends [36]. The molecular chain entanglement between SEPS and these two phases during the blending modification increases the physical compatibility of the blends. With the interfusing of 10 phr SEPS-g-GMA with different grafting ratios, the particle size of dispersed phase in the blends decreases continuously, and the dispersion effect reaches the best state when the grafting ratio of SEPS-g-GMA is 2.44% which is obtained from Fig. 6(d). This may be due to the reaction of SEPS-g-GMA with the carboxyl groups generated by the degradation of the phases in the blends, which promotes the compatibility of the blends. Compatibilizer is similar to the role of emulsifier in emulsion system, which can be beneficial to form cross-linking interface between R-PP and R-HIPS, and enhance the interfacial bonding between two phases in the process of reactive blending. When the grafting ratio of SEPS-g-GMA

is more than 2.44%, the holes in the section change from circular to oval, and the dispersion effect begins to decrease, which can be observed from Fig. 6(e) and (f). This is probably because the grafting rate is too high to exceed the content of carboxyl groups produced by the aging degradation of waste plastics, resulting in a negative effect on the compatibility of the blends. The morphology change of the blend with compatibilizer is consistent with the changing processes of notched impact strength, which improves the compatibilization of the SEPS-g-GMA copolymer with proper grafting ratio in the blends.

3.6 Rheological properties

Fig. 7 shows the effect of SEPS-g-GMA copolymers with different grafting ratios on the complex viscosity of R-PP/R-HIPS blends. The complex viscosity of all formulations shows a downward trend with increasing the frequency, reflecting the shear thinning phenomenon and exhibiting the typical motion characteristics of pseudoplastic fluid [19]. This is because the molecular chain entanglement decreases with increasing the shear force, leading to a better fluidity of blends. However, compared with pure R-PP/R-HIPS blends, the complex viscosity of the blends increases after adding SEPS and SEPS-g-GMA copolymers which is probably due to the physical compatibilization of SEPS with the blends and the reaction between the SEPS-g-GMA and the blends during melt extrusion. But when the grafting ratio of SEPS-g-GMA is more than 2.44%, the high polarity is not conducive to the improved compatibility of the blends, which reduces the entanglement of the chain segments and leads to the decreasing of the complex viscosity.

The relation curves of storage modulus (G') and loss modulus (G'') of R-PP/R-HIPS blends as a function of the frequency are shown in Fig. 8(a) and (b), respectively. The G' and G'' of all samples are observed to increase with increasing the frequency. In the low frequency region, the movement of molecular chain is synchronized with the change of external force, and the flexibility and elasticity of molecular chain are high, which result in a lower G' of the blends. Meanwhile, the smaller friction between the molecules causes the lower G'' . In the high frequency region, the movement of molecular chain could not keep up with the change of external force, and the friction consumption between the molecular chains is also increased, which leads to the increase of G' and G'' [19, 37]. It can also be seen that G' and G'' of the composites with SEPS and SEPS-g-GMA copolymers are higher than those of the pure R-PP/R-HIPS blends, which is due to the chain entanglement between SEPS and the blends, and the chemical reaction of carboxyl groups in the recycled materials with epoxy group in the SEPS-g-GMA. The G' and G'' of the blends reach the maximum value at a graft ratio of SEPS being 2.44%, and the reason may be that the chemical reaction of SEPS-g-GMA and waste materials is more complete. When the grafting ratio is higher, the high polarity improves the interfacial tension of the blends and reduces the ability of chain entanglement, resulting in a lower G' and G'' .

3.7 Thermogravimetric analysis (TGA)

Fig. 9 shows the effect of SEPS-g-GMA copolymers with different grafting ratios on the thermal stability of R-PP/R-HIPS blends. Fig. 9(a) reveals the thermal weight loss curve of the

samples. The thermal weight loss step around 350°C ("m") is probably caused by the degradation of polyethylene (PE). Because the PP used for electrical appliance is generally copolymerized PP with a small amount of PE block. The degradation of the main part of R-PP/R-HIPS blends occurs around 380-470 °C ("n"). Fig. 9(b) is amplified by the "k" region in Fig. 9(a), which corresponds to the beginning degradation of the blends. It can be seen that the initial degradation temperature of the pure blends is the lowest, which may be caused by the reduced molecular weight and poor compatibility of the waste plastics. The waste plastics produced many active sites during the long-term aging process, and the compatibility between the two phases was poor, which made the degradation activation energy of the blends reduced [38]. With the addition of SEPS-g-GMA, the initial degradation temperature of the blends increases significantly and reaches the maximum when the grafting ratio of SEPS-g-GMA is 2.44%. This may be due to the reaction between the epoxy group of SEPS-g-GMA and the carboxyl group of waste plastics, which increases the compatibility and the interfacial bonding between the two phases, thus improving the thermal stability of the blends [39, 40]. However, when the grafting ratio of SEPS-g-GMA is over 2.44%, the initial degradation temperature of blends decreases. This may be because a higher grafting ratio will cause a higher polarity, resulting in an increased interfacial tension, poor compatibility and thermal stability of blends. When the temperature exceeds 480°C, the residual amount in all samples is about 15 wt%, which may be due to the inclusion of calcium carbonate or talcum powders in the waste plastics.

The change of weight loss rate with time is also studied, which can be seen in Fig. 9(c). The corresponding peak at "m" in the derivative thermogravimetric (DTA) curves is probably

derived from the maximum thermal weight loss rate ($T_{\max}$) of polyethylene block in copolymerized polypropylene. In order to analyze the relationship between $T_{\max}$ of the blends and different grafting ratios of SEPS-g-GMA, Fig. 9(d) is obtained by the amplification of region "k" in Fig. 9(c). The $T_{\max}$ of pure R-PP/R-HIPS blends is minimal. With the addition of SEPS-g-GMA, the value of $T_{\max}$ is obviously increased and reaches the maximum when the grafting ratio of SEPS-g-GMA is 2.44%. This may be because the reaction between SEPS-g-GMA and waste plastics improves the compatibility of R-PP/R-HIPS blends, thus inhibiting the degradation [41, 42]. Nevertheless, the higher grafting ratio do not have a positive effect on the stability of blends, which may be attributed to higher polarity which increases the surface tension of the blend interface and decreases the interfacial adhesion.

4. Conclusion

In this work, the SEPS-g-GMA copolymers were prepared by melt grafting in the mixer. The FT-IR result indicated that GMA was successfully grafted on the molecular chain of SEPS. The existence of NVP could promote the graft ratio of SEPS-g-GMA, which was calculated by the acid-base titration. Then, the R-PP/R-HIPS blends with SEPS-g-GMA compatibilizer, prepared by melt blending extrusion and injection molding, were characterized by a variety of testing methods. The FT-IR spectra showed that the epoxy group in SEPS-g-GMA had reacted with the carboxyl group produced by aging and degradation in recycled materials, which improved the chemical compatibilization of SEPS-g-GMA on the blends. The increases of both notch impact strength and elongation at break were probably caused by the physical and

chemical compatibilization of SEPS-g-GMA to the blends. Moreover, the decreasing of particle size and uniform dispersion of the dispersed phase indicated that the interfacial tension between two phases decreased and the bond strength was enhanced with introducing the function of SEPS-g-GMA. Further, the presence of SEPS-g-GMA may improve the chemical compatibility of R-PP/R-HIPS blends, which directly leads to the increase of composite viscosity, storage modulus and loss modulus of the blends. Additionally, the reaction between SEPS-g-GMA and recycled plastics increased the compatibility of R-PP/R-HIPS blends, resulting in a much higher thermal stability. However, a higher grafting ratio would give a higher polarity, which had a negative effect on the compatibility of the blends. The enhanced properties of R-PP/R-HIPS blends obtained under the compatibilization of SEPS-g-GMA expanded the high value recovery and reutilization of R-PP and R-HIPS to obtain functional polymer nanocomposites [43-49] for different potential applications including functional materials, sensing, energy storage [50-68] with properly using different nanofillers.

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References

- [1] V. Pérez-Belis, M.D. Bovea, A. Gómez, Waste electric and electronic toys: Management practices and characterisation, *Resour. Conserv. Recy.* 77 (2013) 1-12.

- [2] J.C. Acomb, M.A. Nahil, P.T. Williams, Thermal processing of plastics from waste electrical and electronic equipment for hydrogen production, *J. Anal. Appl. Pyrol.* 103 (2013) 320-327.
- [3] D. Mahanta, S.A. Dayanidhi, S. Mohanty, S.K. Nayak, Mechanical, thermal, and morphological properties of recycled polycarbonate/recycled poly(acrylonitrile-butadiene-styrene) blend nanocomposites, *Polym. Composite.* 33 (2012) 2114-2124.
- [4] C. Muhammad, J.A. Onwudili, P.T. Williams, Catalytic pyrolysis of waste plastic from electrical and electronic equipment, *J Anal Appl Pyrol.* 113 (2015) 332-339.
- [5] J. Beigbeder, D. Perrin, J.F. Mascaro, J.M. Lopez-Cuesta, Study of the physico-chemical properties of recycled polymers from waste electrical and electronic equipment (WEEE) worked by high resolution near infrared devices, *Resour. Conserv. Recy.* 78 (2013) 105-114.
- [6] K. Vadoudi, J. Kim, B. Laratte, S.J. Lee, N. Troussier, E-waste management and resources recovery in France, *Waste. Manage. Res.* 33 (2015) 919-929.
- [7] P.A. Wäger, R. Hischer, Life cycle assessment of post-consumer plastics production from waste electrical and electronic equipment (WEEE) treatment residues in a Central European plastics recycling plant, *Sci. Total. Environ.* 529 (2015) 158-167.
- [8] P.A. Tarantili, A.N. Mitsakaki, M.A. Petoussi, Processing and properties of engineering plastics recycled from waste electrical and electronic equipment (WEEE), *Polym. Degrad. Stabil.* 95 (2010) 405-410.
- [9] V. Ramesh, M. Biswal, S. Mohanty, S.K. Nayak, Recycling of engineering plastics from waste electrical and electronic equipments: influence of virgin polycarbonate and impact modifier on the final performance of blends, *Waste. Manage. Res.* 32 (2014) 379-88.
- [10] D. Perrin, O. Mantaux, P. Ienny, R. Leger, M. Dumon, J.M. Lopez, Influence of impurities on the performances of HIPS recycled from Waste Electric and Electronic Equipment

- (WEEE), *Waste. Manage.* 56 (2016) 438-445.
- [11] M. Benedetti, L. Cafiero, D. De Angelis, A. Dell'Era, M. Pasquali, S. Stendardo, R. Tuffi, S.V. Cipriotti, Pyrolysis of WEEE plastics using catalysts produced from fly ash of coal gasification, *Front. Env. Sci. Eng.* 11 (2017) 11.
- [12] H. Kang, Q. Shao, X. Guo, A. Galaska, Y. Liu, Z. Guo, Separation and recovery of copper foil and fabric from waste printed circuit boards by decomposing brominated epoxy resin using near-critical water, *Eng. Sci.*, (2018), in press, <https://doi.org/10.30919/espun.es.180312>.
- [13] Z. Wang, N.J Miles, T. Wu, F. Gu, P. Hall, Recycling oriented vertical vibratory separation of copper and polypropylene particles, *Powder. Technol.* 301 (2016) 694-700.
- [14] J. Zhao, B.A. Li, X. Li, Y.C. Qin, C. Li, S.C. Wang, Numerical simulation of novel polypropylene hollow fiber heat exchanger and analysis of its characteristics, *Appl. Therm. Eng.* 59 (2013) 134-141.
- [15] Y.V. Vazquez, S.E. Barbosa, Compatibilization Strategies for Recycling Applications of High Impact Polystyrene/Acrylonitrile Butadiene Blends, *J. Polym. Environ.* (2016) 1-10.
- [16] S.R. Mallampati, B.H. Lee, Y. Mitoma, C. Simion, Selective sequential separation of ABS/HIPS and PVC from automobile and electronic waste shredder residue by hybrid nano-Fe/Ca/CaO assisted ozonisation process, *Waste. Manage.* 60 (2017) 428-438.
- [17] Y.V. Vazquez, S.E. Barbosa, Process Window for Direct Recycling of Acrylonitrile-Butadiene-Styrene and High-Impact Polystyrene from Electrical and Electronic Equipment Waste, *Waste.Manage.* 59 (2017) 403-408.
- [18] Y.C. Li, X.L. Wu, J.F. Song, J.F. Li, Q. Shao, N. Cao, N. Lu, Z.H. Guo, Repurification of recycled acrylonitrile-butadiene-styrene by pyromellitic dianhydride: Repurification performance evaluation and property analysis, *Polymer.* 124 (2017) 41-47.

- [19] J. Wang, Y.C. Li, J.F. Song, M.Y. He, J.J. Song, K. Xia, Recycling of acrylonitrile-butadiene-styrene (ABS) copolymers from waste electrical and electronic equipment (WEEE), through using an epoxy-based chain extender, *Polym. Degrad. Stabil.* 112 (2015) 167-174.
- [20] B. Qiao, G.S. Hu, M.Y. He, Y.C. Li, Reactive Compatibilization of Recycled HIPS/ABS Blends with Maleic-Anhydride Grafted Styrene-Ethylene-Butadiene-Styrene and Oxazoline-Modified Recycled ABS, *Oxid. Commun.* 39 (2016) 2085-2100.
- [21] S.S. Wang, N. Wang, L. Meng, J. Zhao, Y. Feng, Multi functionalization of polypropylene with controlled degradation and its structure characterization, *Macromol. Res.* 19 (2011) 951-964.
- [22] N. Ayilimis, M. Tasdemir, T. Akbulut, Water absorption and mechanical properties of PP/HIPS hybrid composites filled with wood flour, *Polym. Composite.* 38 (2015) 863-869.
- [23] N. Kawahara, J. Saito, S. Matsuo, H. Kaneko, T. Matsugi, S. Kojoh, N. Kashiwa, New methodology for synthesizing polypropylene-graft-polystyrene (PP-g-PS) by coupling reaction with brominated polypropylene, *Polym. Bull.* 59 (2007) 177-183.
- [24] J. Uribe-Calderon, C. Díaz-Arriaga, The effects of carbon nanotubes, blend composition and glycidyl methacrylate-grafted polypropylene compatibilizer on the morphology, mechanical and electrical properties of polypropylene-polyamide 6 blends, *Polym. Bull.* 74 (2016) 1573-1593.
- [25] M. Studzinski, R. Jeziorska, A. Szadkowska, M. Zielecka, Modified nanosilica-filled polypropylene composites with glycidyl methacrylate grafted ethylene/n-octene copolymer as compatibilizer, *Polimery-W*, 59 (2014) 625-635.
- [26] O. Jazani, H. Rastin, K. Formela, A. Hejna, M. Shahbazi, B. Farkiani, M. Saeb, An investigation on the role of GMA grafting degree on the efficiency of PET/PP-g-GMA reactive blending: morphology and mechanical properties, *Polym. Bull.* 74 (2017)

4483-4497.

- [27] H.W. Xiao, F.Y. Yu, Y. Yu, S.Q. Huang, Grafting of glycidyl methylacrylate onto chlorinated polypropylene and its bonding to aluminum flake, *J. Appl. Polym. Sci.* 104 (2007) 2515-2521.
- [28] H.W. Yang, S.F. Luan, J. Zhao, H.C. Shi, Q.A. Shi, J.H. Yin, P. Stagnaro, N-vinyl pyrrolidone-assisted free radical functionalization of glycidyl methacrylate onto styrene-b-(ethylene-co-butylene)-b-styrene, *React. Funct. Polym.* 70 (2010) 961-966.
- [29] S.F. Luan, J. Zhao, H.W. Yang, H.C. Shi, J. Jin, X.M. Li, J.C. Liu, J.W. Wang, J.H. Yin, P. Stagnaro, Surface modification of poly(styrene-b-(ethylene-co-butylene)-b-styrene) elastomer via UV-induced graft polymerization of N-vinyl pyrrolidone, *Colloid. Surface. B.* 93 (2012) 127-134.
- [30] X. Jiang, D.X. Zhou, X.L. Huang, W.F. Zhao, C.S. Zhao, Hexanediamine functionalized poly(glycidyl methacrylate-co-N-vinylpyrrolidone) particles for bilirubin removal, *J. Colloid. Interf. Sci.* 504 (2017) 214-222.
- [31] J.L. Li, X.M. Xie, Reconsideration on the mechanism of free-radical melt grafting of glycidyl methacrylate on polyolefin, *Polymer*, 53 (2012) 2197-2204.
- [32] A. Kumar, T.V. Rao, S.R. Chowdhury, S.V.S.R. Reddy, Compatibility confirmation and refinement of thermal and mechanical properties of poly(lactic acid)/poly(ethylene-co-glycidyl methacrylate) blend reinforced by hexagonal boron nitride, *React. Funct. Polym.* 117 (2017) 1-9.
- [33] J. Parameswaranpillai, G. Joseph, K.P. Shinu, K.P. Shinu, S. Jose, N.V. Salim, N. Hameed, Development of hybrid composites for automotive applications: Effect of addition of SEBS on the morphology, mechanical, viscoelastic, crystallization and thermal degradation properties of PP/PS-xGnP composites, *RSC. Adv.* 5 (2015) 25634-25641.
- [34] H.R. Li, J.W. Wang, G.H. Li, Y.Q. Lu, N.Y. Wang, Q.X. Zhang, X.W. Qu, Preparation of

- core-shell structured particle and its application in toughening PA6/PBT blends, *Polym. Advan. Technol.* 28 (2017) 699-707.
- [35] X. Li, H.L. Kang, J.X. Shen, L.Q. Zhang, T. Nishi, K. Ito, C.M. Zhao, P. Coates, Highly toughened polylactide with novel sliding graft copolymer by in situ reactive compatibilization, crosslinking and chain extension, *Polymer*. 55 (2014) 4313-4323.
- [36] R.Q. Chen, X.L. Jiang, F. You, C. Yao, Optimizing the morphology, mechanical and crystal properties of in-situ polypropylene/polystyrene blends by reactive extrusion, *Fiber. Polym.* 17 (2016) 1550-1557.
- [37] R. Dhavalikar, M. Yamaguchi, M. Xanthos, Molecular and structural analysis of a triepoxide-modified poly(ethylene terephthalate) from rheological data, *J. Polym. Sci. Pol. Chem.* 41 (2003) 958-969.
- [38] B. Sun, D.K. Qin, G.Z. Liang, A.G. Gu, L. Yuan, The relationship between the compatibility and thermodegradation stability of modified polyetherimide/bismaleimide resins by hyperbranched polysiloxane with high degree of branching, *Polym. Advan. Technol.* 24 (2013) 1051-1061.
- [39] M. De Guzman, P.Y. Liu, J.T. Chen, K.L. Tung, K.R. Lee, J.Y. Lai, Effect of compatibilizer on compatibility and pervaporation performance of PC/PHEMA blend membranes, *J. Membrane. Sci.* 378 (2011) 503-511.
- [40] I. Krupa, G. Miková, A.S. Luyt, Polypropylene as a potential matrix for the creation of shape stabilized phase change materials, *Eur. Polym. J.* 43 (2007) 895-907.
- [41] A.B. Nair, P. Kurian, R. Joseph, Ethylene-propylene-diene terpolymer/hexa fluoropropylene-vinylidene fluoride dipolymer rubber blends: Thermal and mechanical properties, *Mater. Design.* 36 (2012) 767-778.
- [42] N. Wang, J.G. Yu, P.R. Chang, X.F. Ma, Influence of formamide and water on the properties of thermoplastic starch/poly(lactic acid) blends, *Carbohydr. Polym.* 71 (2008) 109-118.

- [43] Y.L. Ma, L. Lv, Y.R. Guo, Y.J. Fu, Q. Shao, T.T. Wu, S.J. Guo, K. Sun, X.K. Guo, E.K. Wujcik, Z.H. Guo, Porous lignin based poly(acrylic acid)/organo-montmorillonite nanocomposites: swelling behaviors and rapid removal of Pb (II) ions, *Polymer*. 128 (2017) 12-23.
- [44] C.F. Wang, M. Zhao, J. Li, J.L. Yu, S.F. Sun, S.S. Ge, X.K. Guo, F. Xie, B. Jiang, E.K. Wujcik, Y.D. Huang, N. Wang, Z.H. Guo, Silver nanoparticles/graphene oxide decorated carbon fiber synergistic reinforcement in epoxy-based composites, *Polymer*. 131 (2017) 263–271.
- [45] J.B. Zhao, L.L. Wu, C.X. Zhan, Q. Shao, Z.H. Guo, L.Q. Zhang, Influence of formamide and water on the properties of thermoplastic starch/poly(lactic acid) blends, *Polymer*. 133 (2017) 272-287.
- [46] P Zhou, S. Wang, C.L. Tao, X.K. Guo, L.H. Hao, Q. Shao, L. Liu, Y.P. Wang, W. Chu, B. Wang, S.Z. Luo, Z.H. Guo, PAA/alumina composites prepared with different molecular weight polymers and utilized as support for nickel-based catalyst, *Adv. Polym. Technol.* (2018), <https://doi.org/10.1002/adv.21908> (in press).
- [47] Y.X. He, S. Yang, H. Liu, Q. Shao, Q.Y. Chen, C. Lu, Y.L. Jiang, C.T. Liu, Z.H. Guo, Influence of formamide and water on the properties of thermoplastic starch/poly(lactic acid) blends, *J. Colloid Interf. Sci.* 517 (2018) 40-51.
- [48] W.Q. Yang, X.L. Wang, J.F. Li, X.R. Yan, S.S. Ge, S. Tadakamalla, Z.H. Guo, Influence of formamide and water on the properties of thermoplastic starch/poly(lactic acid) blends, *Polym. Eng. Sci.* (2018), <https://doi.org/10.1002/pen.24675> (in press).
- [49] X.K. Cui, G.Y. Zhu, Y.F. Pan, Q. Shao, C. Zhao, M.Y. Dong, Y. Zhang, Z.H. Guo, Polydimethylsiloxane-titania nanocomposite coating: fabrication and corrosion resistance, *Polymer*. 138 (2018) 203-210.
- [50] L. Zhang, W. Yu, C. Han, J. Guo, Q.H. Zhang, H.Y. Xie, Q. Shao, Z.G. Sun, Z.H. Guo,

- Large scaled synthesis of heterostructured electrospun TiO₂/SnO₂ nanofibers with an enhanced photocatalytic activity, *J. Electrochem. Soc.* 164 (2017), H651-H656.
- [51] L. Zhang, M.K. Qin, W. Yu, Q.H. Zhang, H.Y. Xie, Z.G. Sun, Q. Shao, X.K. Guo, L.H. Hao, Y.J. Zheng, Z.H. Guo, Heterostructured TiO₂/WO₃ nanocomposites for photocatalytic degradation of toluene under visible light, *J. Electrochemical Soc.* 164 (2017) H1086-H1090.
- [52] B. Song, T.T. Wang, H.G. Sun, Q. Shao, J.K. Zhao, K.K. Song, L.H. Hao, L. Wang, Z.H. Guo, Two-step hydrothermally synthesized carbon nanodots/WO₃ photocatalysts with enhanced photocatalytic performance, *Dalton Trans.* 46 (2017) 15769-15777.
- [53] J.N. Huang, Y.H. Cao, J. Shao, X.F. Peng, Z.H. Guo, Magnetic nanocarbon adsorbents with enhanced hexavalent chromium removal: morphology dependence of fibrillar vs particulate structures, *Ind. Eng. Chem. Res.* 56 (2017) 10689–10701.
- [54] K.D. Gong, Q. Hu, L. Yao, M. Li, D.Z. Sun, Q. Shao, B. Qiu, Z.H. Guo, Ultrasonic pretreated sludge derived stable magnetic active carbon for Cr(VI) removal from wastewater, *ACS Sustain. Chem. Eng.* (2018), <https://doi.org/10.1021/acssuschemeng.7b04421> (in press).
- [55] C.B. Cheng, R.H. Fan, Z.Y. Wang, Q. Shao, X.K. Guo, P.T. Xie, Y.S. Yin, Y.L. Zhang, L.Q. An, Y.H. Lei, J.E. Ryu, A. Shankar, Z.H. Guo, Tunable and weakly negative permittivity in carbon/silicon nitride composites with different carbonizing temperatures, *Carbon*. 125 (2017) 103-112.
- [56] Z. Hu, Q. Shao, Y. Huang, L. Yu, D. Zhang, X. Xu, J. Lin, H. Liu, Z. Guo, Light triggered interfacial damage self-healing of poly(p- phenylene benzobisoxazole) fiber composites, *Nanotechnology*, 29 (2018) 185602.
- [57] K. Sun, R. Fan, X. Zhang, Z. Zhang, Z. Shi, N. Wang, P. Xie, Z. Wang, G. Fan, H. Liu, C. Liu, T. Li, C. Yan and Z. Guo An overview of metamaterials and their achievements in

- wireless power transfer, *J. Mater. Chem. C*, 6 (2018), 2925-2943.
- [58] Y. Guo, G. Xu, X. Yang, K. Ruan, T. Ma, Q. Zhang, J. Gu, Y. Wu, H. Liu and Z. Guo, Significantly enhanced and precisely modeled thermal conductivity in polyimide nanocomposites by chemically modified graphene via in-situ polymerization and electrospinning-hot press technology, *J. Mater. Chem. C*, 6 (2018) 3004-3015.
- [59] X. Wang, X. Liu, H. Yuan, et al., Non-covalently functionalized graphene strengthened poly(vinyl alcohol), *Mater. Design*, 139 (2018) 372-379.
- [60] J. Lin, X. Chen, C. Chen, J. Hu, C. Zhou, X. Cai, W. Wang, C. Zheng, R. Zhang, J. Cheng, H. Liu, and Z. Guo, Durably antibacterial and bacterially anti-adhesive cotton fabrics coated by cationic fluorinated polymers, *ACS Appl. Mater. Interfaces*, 10 (2018) 6124-6136.
- [61] Y. Li, B. Zhou, G. Zheng, X. Liu, T. Li, C. Yan, C. Cheng, K. Dai, C. Liu, C. Shen and Z. Guo, Continuously prepared highly conductive and stretchable SWNTs/MWNTs synergistically composited electrospun thermoplastic polyurethane yarns for wearable sensing, *J. Mater. Chem. C*, 6 (2018) 2258-2269.
- [62] X. Guan, G. Zheng, K. Dai, et al., Carbon nanotubes-adsorbed electrospun PA66 nanofiber bundles with improved conductivity and robust flexibility, *ACS Appl. Mater. Interfaces*, 8 (2016) 14150-14159.
- [63] H. Liu, W. Huang, X. Yang, et al., Organic vapor sensing behaviors of conductive thermoplastic polyurethane-graphene nanocomposites, *J. Mater. Chem. C*, 4 (2016) 4459-4469.
- [64] H. Liu, M. Dong, W. Huang, et al., Lightweight conductive graphene/thermoplastic polyurethane foams with ultrahigh compressibility for piezoresistive sensing, *J. Mater. Chem. C*, 5 (2017) 73-83.
- [65] H. Liu, Y. Li, K. Dai, et al., Electrically conductive thermoplastic elastomer nanocomposites at ultralow graphene loading levels for strain sensor applications, *J. Mater. Chem. C*, 4

(2016) 157-166.

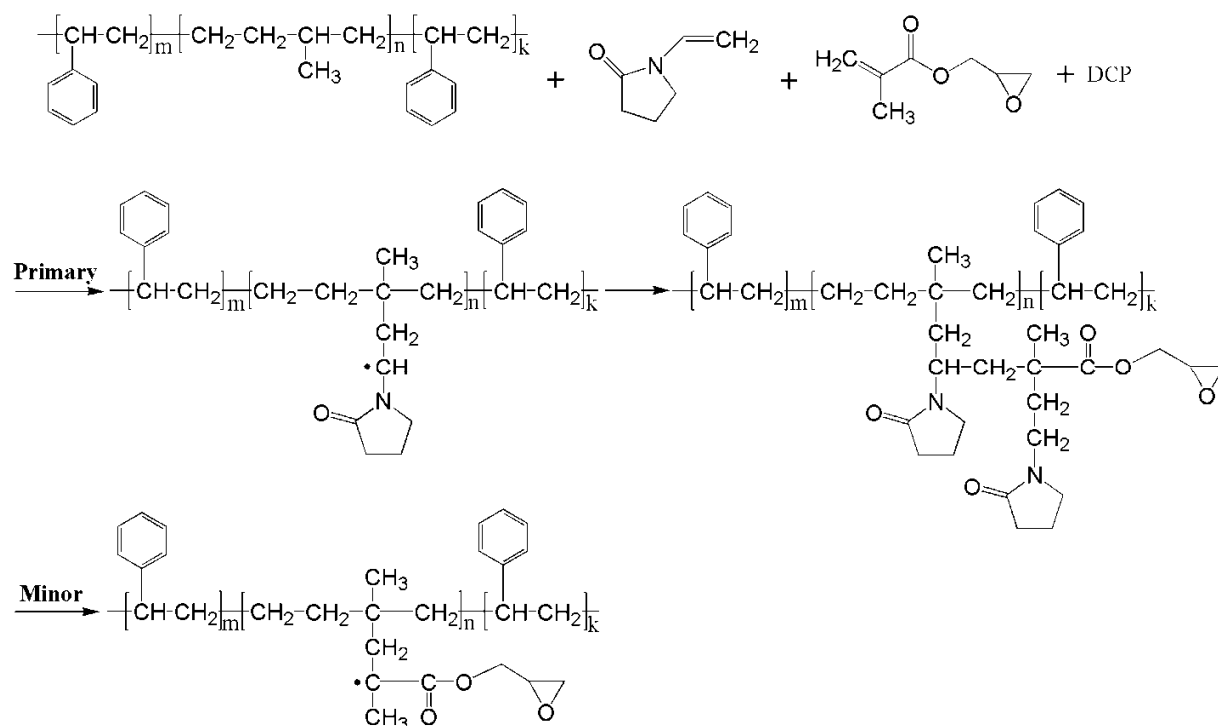
- [66] Z. Hu, C. Wang, F. Zhao, et al. Fabrication of a graphene/C60 nanohybrid via gamma-cyclodextrin host-guest chemistry for photodynamic and photothermal therapy. *Nanoscale* 9 (2017) 8825.
- [67] J. Gu, W. Dong, Y. Tang, et al. Ultra-low Dielectric, Fluoride-containing Cyanate Ester Resins Combining with Prominent Mechanical Properties and Excellent Thermal and Dimension Stabilities. *J. Mater. Chem. C*, 5 (2017) 6929-6936.
- [68] X. Duan, T. Bai, J. Du and J. Kong. One-pot synthesis of redox-reduction responsive amphiphilic drug self-delivery micelle of doxorubicin-disulfide-methoxy polyethylene glycol for tumor therapy, *J. Mater. Chem. B*, 6 (2018) 39-43.

Table 1. The accurate formulations for the synthesis of SEPS-g-GMA.

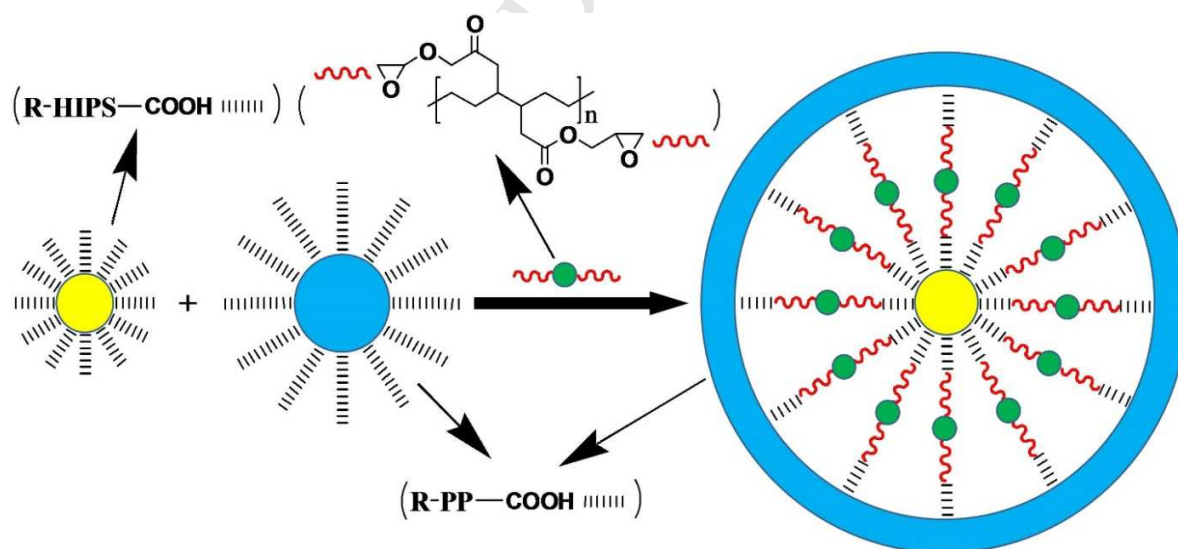
Sample	SEPS/phr	DCP/phr	GMA/phr	NVP/phr
S1	100	0	0	0
S2	100	0.4	6.0	4.0
S3	100	0.4	6.0	5.0
S4	100	0.4	6.0	6.0
S5	100	0.4	6.0	7.0

Table 2 The experimental formulas of R-PP/R-HIPS/SEPS-g-GMA blends.

Sample	R-PP/phr	R-HIPS/phr	SEPS/phr	SEPS-g-GMA/phr
B1	70	30	0	0
B2	70	30	10	0
B3	70	30	0	10 (G=1.10%)
B4	70	30	0	10 (G=2.44%)
B5	70	30	0	10 (G=4.40%)
B6	70	30	0	10 (G=5.40%)



Scheme 1. The mechanism of the preparation of SEPS-g-GMA copolymer.



Scheme 2. The mechanism of the chemical reaction between SEPS-g-GMA and R-PP/R-HIPS blends.

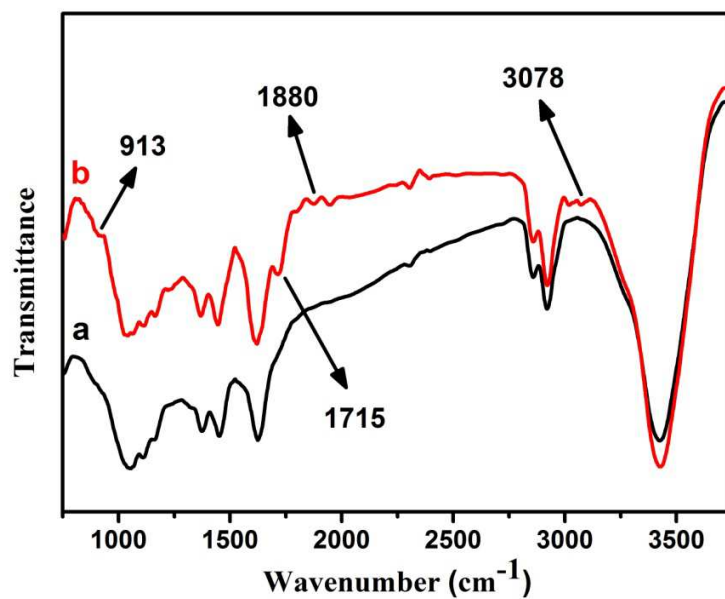


Fig. 1. FT-IR spectra of SEPS (a) and SEPS-g-GMA (b).

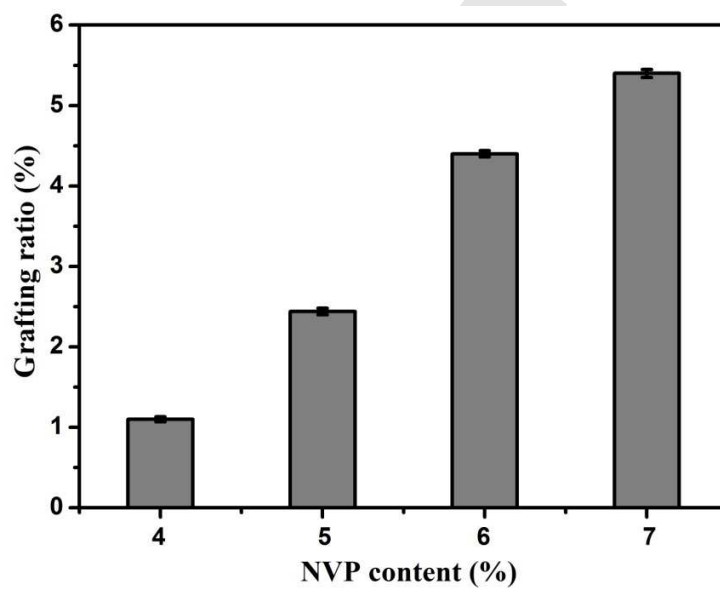


Fig. 2. The effect of the content of NVP on the grafting ratio of SEPS-g-GMA.

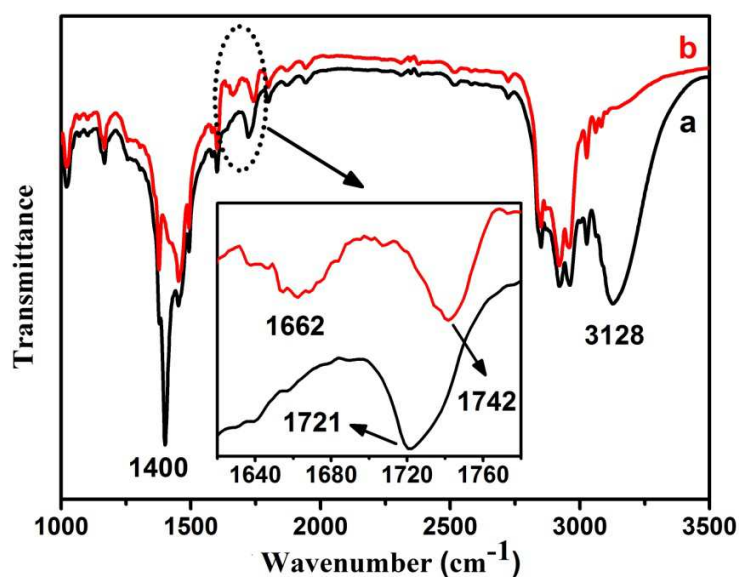


Fig. 3. FT-IR spectra of R-PP/R-HIPS blends (a) and R-PP/R-HIPS/SEPS-g-GMA blends (b).

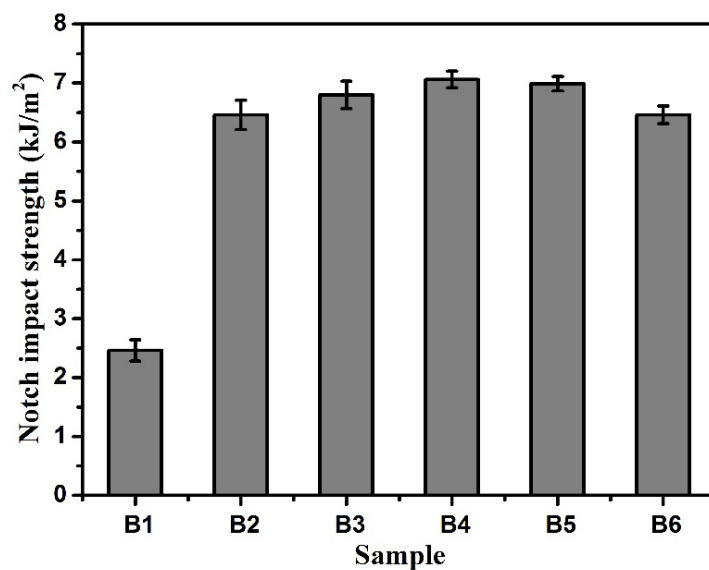


Fig. 4. Effect of SEPS-g-GMA with different grafting ratios on the notch impact strength of the R-PP/R-HIPS blends.

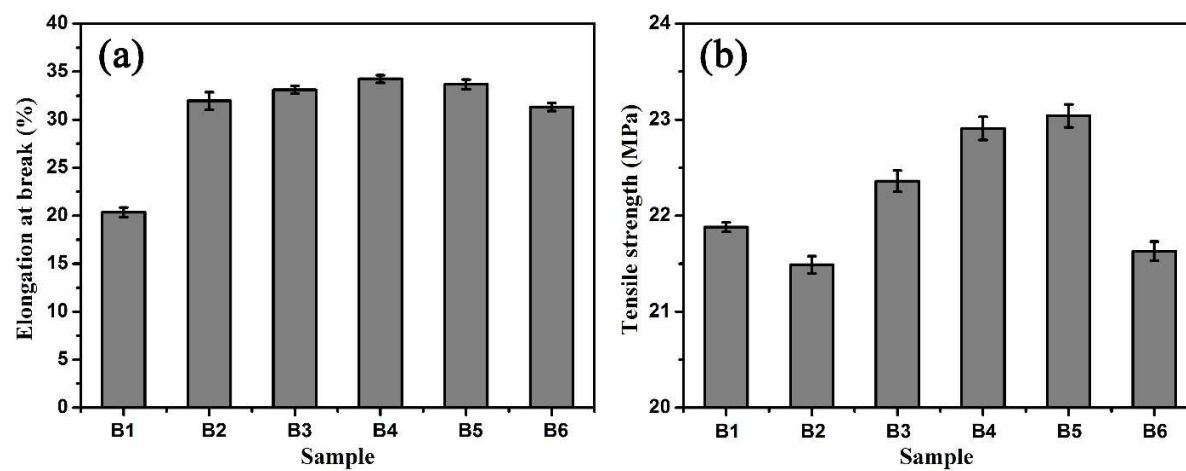


Fig. 5. Effect of SEPS-g-GMA with different grafting ratios on elongation at break and tensile strength of the R-PP/R-HIPS blends.

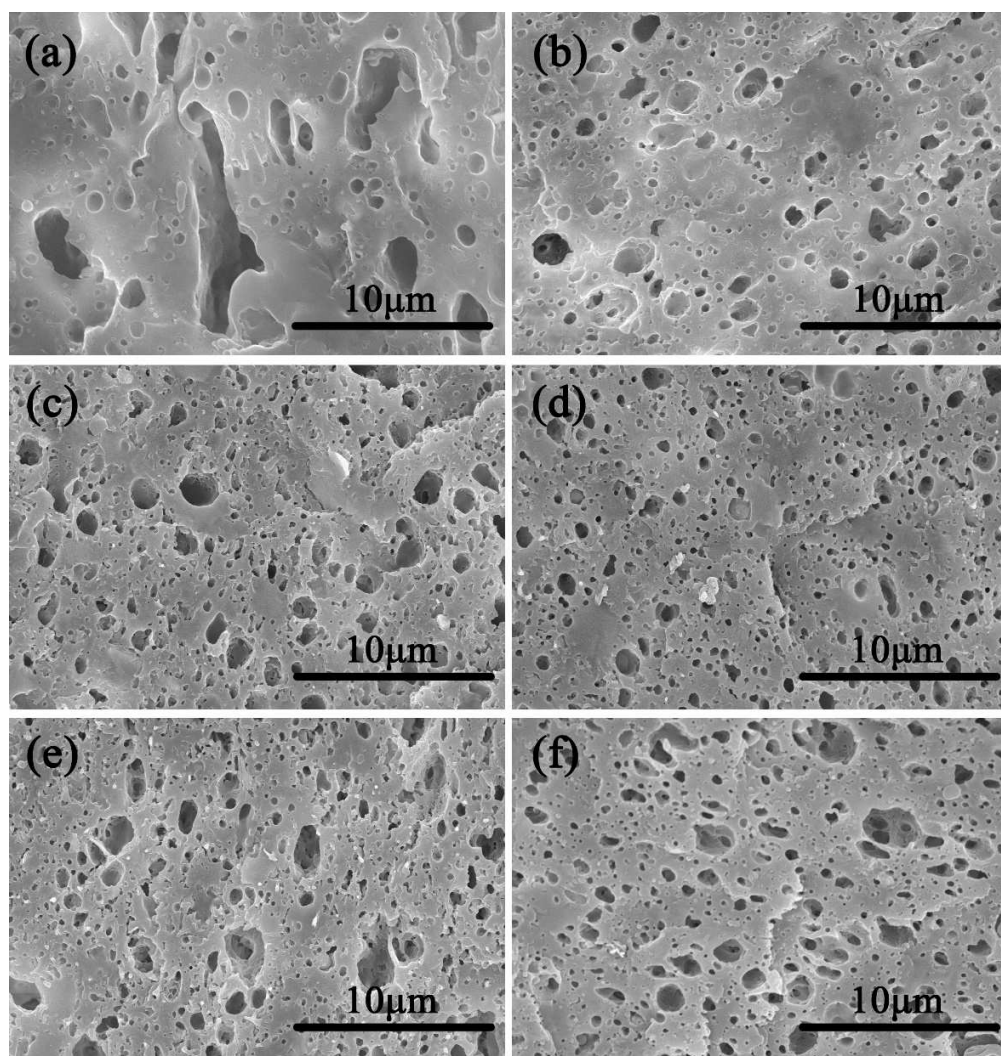


Fig. 6. Effect of SEPS-g-GMA with different grafting ratios on the morphology of the R-PP/R-HIPS blends.

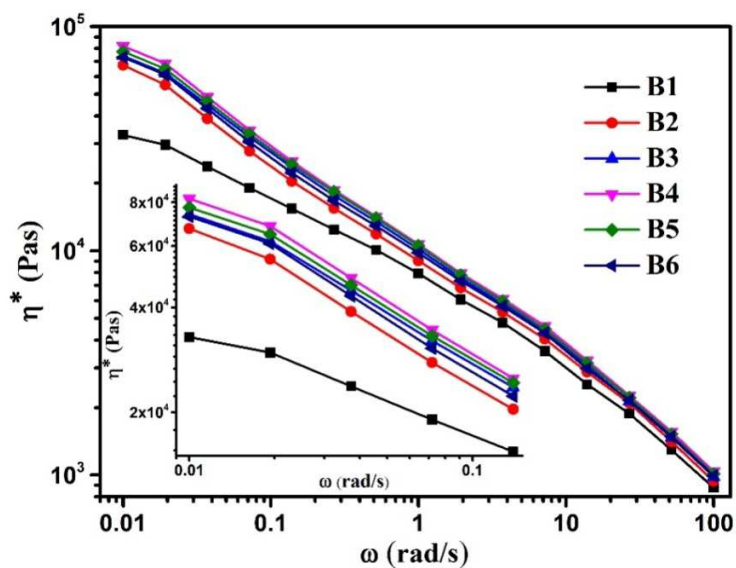


Fig. 7. Effect of SEPS-g-GMA with different grafting ratios on the complex viscosity of the R-PP/R-HIPS blends.

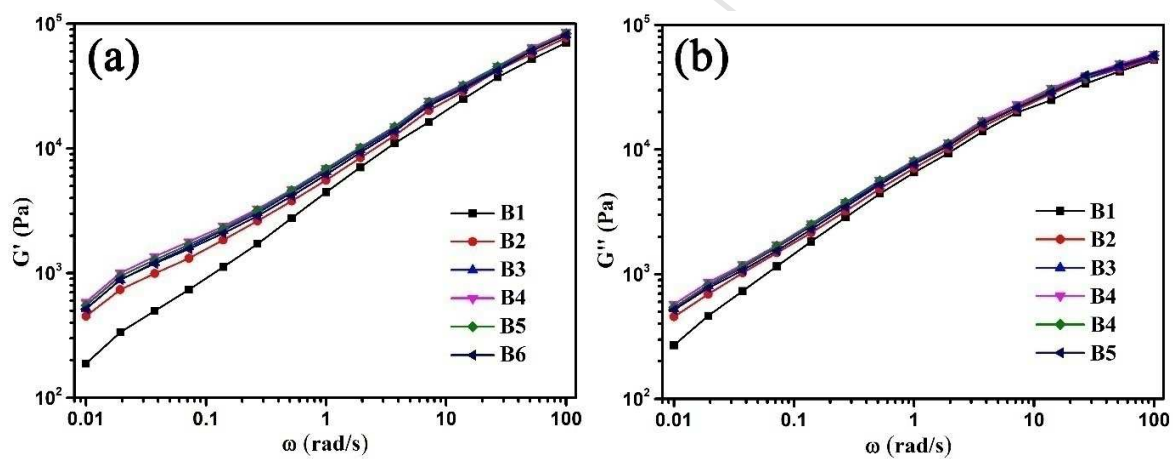


Fig. 8. The relation curves of the storage modulus and loss modulus of R-PP/R-HIPS blends with the frequency variation.

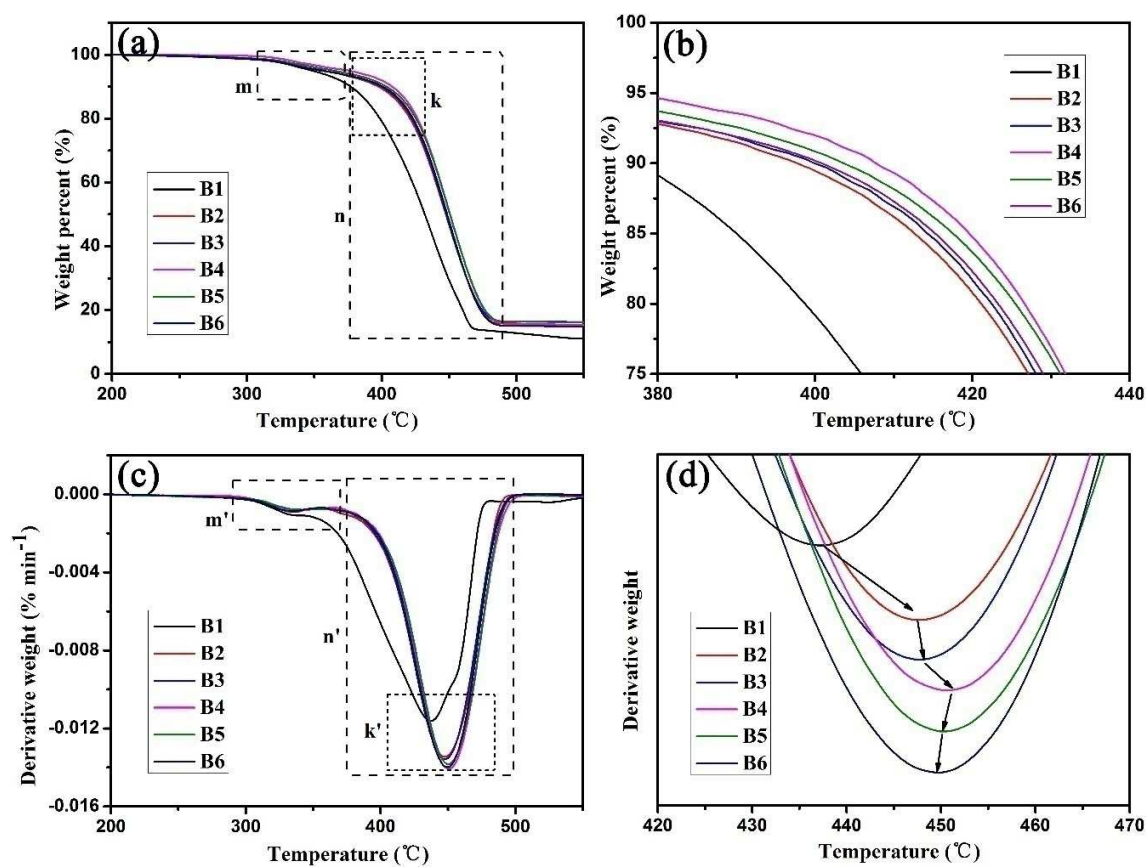


Fig. 9. Effect of SEPS-g-GMA with different grafting ratios on the thermal stability of the R-PP/R-HIPS blends.

Highlights

- SEPS-g-GMA was successfully prepared and the effect of NVP content on the grafting ratio of SEPS-g-GMA was tested.
- The notch impact strength of R-PP/R-HIPS blends was significantly improved by 186.99% when the SEPS-g-GMA(G=2.44%) content was 10 wt%.
- The compatibilization mechanism of SEPS-g-GMA to R-PP/R-HIPS blends was explained.
- The SEPS-g-GMA compatibilization effect on R-PP/R-HIPS blends was interpreted.